

Ouranos Reconstruction – RDPS-CaPA (ORRC) v1.0 / Reconstruction Ouranos – RDPS-CAPA (RORC) v1.0

Asli Bese and Travis Logan

Ouranos Inc., 550 Rue Sherbrooke W., West Tower, 19th floor,
Montreal, QC, Canada H3A 1B9

March 24, 2026

1 Background and Summary

The Canadian Surface Reanalysis (CaSR) dataset is a high-resolution reanalysis product developed by Environment and Climate Change Canada (ECCC) and provides an observation-based representation of surface and near-surface atmospheric conditions over North America, Central America, and nearly all of Greenland (Gasset et al., 2021). CaSR version 2.1 covers 1980–2018, and CaSR version 3.2 extends the record to 1980–2024. Continuous climate monitoring and operational climate services, however, require high-resolution reanalysis data through the present day. To ensure continuity beyond the CaSR period, we produced the Ouranos Reconstruction – RDPS-CaPA (ORRC), version 1.0. The methodological framework was designed by ECCC to generate a CaSR-like dataset from the Regional Deterministic Prediction System (RDPS) and the Canadian Precipitation Analysis (CaPA), and Ouranos implemented the end-to-end data production. ORRC v1.0 provides hourly surface and near-surface meteorological fields over the same domain as CaSR v3.2 at a resolution of 0.09° (~10 km) on a rotated uniform latitude-longitude grid, over land and ocean, from 2015 to present.

ORRC's primary contribution is temporal continuity at high spatial and temporal resolution, enabling near-real-time climate monitoring and applications that compare the most recent conditions against historical baselines. This is particularly valuable for operational climate services, event monitoring, and routinely updating climate indicators without waiting for reanalysis releases.

2 Methodology

2.1 Data

2.1.1 Regional Deterministic Prediction System (RDPS)

RDPS is ECCC's operational short-range deterministic numerical weather prediction system, based on a limited-area configuration of the Global Environmental Multiscale (GEM) model (ECCC, 2025). It produces forecasts of near-surface meteorological variables, including temperature, precipitation, cloud cover, wind speed and direction, humidity and others. Forecasts extend to 84 hours and are issued four times per day (at 00, 06, 12, and 18 UTC) over Canada and the United States at ~10 km horizontal resolution. To construct ORRC, we retain only the 6- to 12-hour forecast lead times from each RDPS cycle to minimize spin-up errors and model drift.

2.1.2 Canadian Precipitation Analysis (CaPA-RDPA)

In its deterministic configuration, CaPA produces the Regional Deterministic Precipitation Analysis (RDPA), providing optimal estimates of precipitation amounts over 6-hour periods (at 00, 06, 12, and 18 UTC) and 24-hour periods (at 06 and 12 UTC) on a ~10 km grid over North America (Canada, United States, Mexico) (Fortin et al., 2018; Lespinas et al., 2015). It combines in situ precipitation gauge observations, weather-radar quantitative precipitation estimates, and a short-term precipitation forecast (trial field) from RDPS using an optimal interpolation (OI) framework. The system applies a variance-stabilizing transformation (cubic root) prior to analysis and updates error statistics through a temporally adaptive approach, alongside quality control of precipitation observations (Fortin et al., 2018; Lespinas et al., 2015). In ORRC production, we use the 6-hour precipitation as the reference accumulation to adjust RDPS precipitation and produce a CaSR-like precipitation field.

2.1.3 Data retrieval

RDPS and CaPA data up to March 31st, 2025 were downloaded in NetCDF format from the Canadian Surface Prediction Archive (CaSPAr) at <https://caspar-data.ca> (Mai et al., 2020) under the ECCC Data Servers End-use Licence (https://eccc-msc.github.io/open-data/licence/readme_en/). Because of CaSPAr service unavailability, RDPS and CaPA data starting from April 1st, 2025 onward, together with CaSR v3.2 data, were downloaded from ECCC's high-performance computer GPSC-C at <https://hpfx.collab.science.gc.ca/> and converted to NetCDF¹.

¹ Note that the Ouranos ORRC v1.0 data treatment workflow retrieves RDPS and CaPA data in ECCC's FST file format via a GPSC-C collaborator account and not the public view of the provided link which exposes data in grib2 format. Conversion to NetCDF is achieved using the `fst2nc` Python utilities: <https://github.com/neishm/fstd2nc>

2.2 Preprocessing

Prior to ORRC production, we perform a completeness and consistency check of the raw RDPS and CaPA data. For RDPS, we verify that each forecast cycle includes the seven hourly lead times (+6 to +12 h). For CaPA, we verify that each 6-hour accumulation period is represented by a file containing a single analysis time step. Missing files are generated and populated with NaN values, and any missing variables in existing files are similarly filled with NaNs.

We regrid the RDPS and CaPA data to the CaSR v3.2 target grid (0.09° resolution) using nearest-neighbour interpolation. Although the input and target grids both have the ~10 km horizontal resolution, variations in grid definitions, including changes to the RDPS operational grid over time, necessitate this regridding step. Nearest-neighbour interpolation is used because it preserves original values and can accommodate varying grid structures.

2.3 ORRC Production

The general methodology for ORRC dataset construction follows that described by ECCC researchers for the construction of [CaSR-like files based on real-time operational products](#) (dimmlena, 2023). The details for the individual steps are described in the following sub-sections.

2.3.1 Temporal assembly and variables

Each RDPS forecast cycle provides hourly fields for lead times +6 to +12. ORRC v1.0 is constructed by concatenating these selected segments from consecutive forecast cycles to form a continuous hourly time series. The resulting 24-hour reporting window spans from 13 UTC to 12 UTC of the following calendar day.

ORRC v1.0 provides an hourly record of near-surface meteorological fields, including near-surface air temperature, dew point temperature, relative humidity, specific humidity, surface air pressure, sea level pressure, wind speed and direction, downwelling shortwave and longwave radiation at the surface, and precipitation (model precipitation and precipitation adjusted with CaPA 6-hour totals).

The temporal coverage of ORRC v1.0 (2015–present) is determined by the availability of the underlying RDPS and CaPA inputs in the CaSPAr archives for the variables used here. The eastward and northward near-surface wind components contain erroneous values in the CaSPAr archive from June 11, 2024 onward, which seems due to a change in grid definition. Accordingly, these variables are set to NaN in ORRC v1.0 from June 11, 2024 until access to

GPSC-C became available on April 1, 2025. Users requiring near-surface wind components during this interval should rely on CaSR v3.2 (1980–2024) or alternative wind products.

We additionally provide daily aggregates, including daily means and daily minima/maxima for near-surface air temperature, dew point temperature, and relative humidity. Daily values are computed from the hourly record by resampling to a daily time step using standard calendar-day bins defined by the timestamps (midnight-to-midnight).

Table 1 shows the list of variables provided in ORRC v1.0, their corresponding ECCC identifiers and output frequencies.

Table 1. Variables provided in ORRC v1.0. The table lists the long name, ORRC variable name, corresponding ECCC variable identifier, and output frequency. Variables marked with * are also available in the bias-adjusted ORRC-a product (daily only).

Long name	Variable name (ORRC)	Variable name (ECCC)	Frequency
Precipitation (with CaPA blending)*	pr	A_PR_SFC	Hourly + daily mean
Precipitation (forecast)	prmod	P_PR_SFC	Hourly + daily mean
1.5 metre temperature*	tas	P_TT_10000	Hourly + daily mean/min/max
1.5 metre dewpoint temperature	tdps	P_TD_10000	Hourly + daily mean/min/max
Near-Surface Relative Humidity (1.5m)	hurs	P_HR_10000	Hourly + daily mean/min/max
Near-Surface Specific Humidity (1.5m)	huss	P_HU_10000	Hourly + daily mean
Surface Downwelling Shortwave Radiation	rsds	P_FB_SFC	Hourly + daily mean
Surface Downwelling Longwave Radiation	rlds	P_FI_SFC	Hourly + daily mean
Surface Air Pressure	ps	P_P0_SFC	Hourly + daily mean
Sea Level Pressure	psl	P_PN_SFC	Hourly + daily mean
Eastward Near-Surface Wind (10m)	uas	P_UUC_10000	Hourly + daily mean
Northward Near-Surface Wind (10m)	vas	P_VVC_10000	Hourly + daily mean
Near-Surface Wind Speed (10m)	sfcWind	P_UVC_10000	Hourly + daily mean
Near-Surface Meteorological Wind Direction (10m)	winddir	P_WDC_10000	Hourly + daily mean

2.3.2 Precipitation blending

To produce a CaSR-like precipitation field, we blend RDPS and CaPA by scaling RDPS hourly precipitation increments so that each 6-hour accumulation matches the corresponding CaPA total. RDPS precipitation is provided as a cumulative field, where each time step represents the total precipitation accumulated from the start of the forecast cycle. We convert cumulative precipitation to hourly increments by differencing consecutive forecast times. Hourly increments are then summed to obtain RDPS 6-hour totals aligned with the CaPA 6-hour accumulation periods.

For each 6-hour window, we compute a grid-cell scaling factor as the ratio of the CaPA precipitation total to the RDPS precipitation total and apply this factor to the corresponding RDPS hourly increments. This ensures that the adjusted hourly precipitation sums to the CaPA 6-hour accumulation while preserving the hourly distribution of precipitation within each window.

To avoid unstable ratios when RDPS totals are near zero, we apply a small threshold of 1×10^{-6} m for the 6-hour accumulated precipitation. Where the RDPS 6-hour total exceeds this threshold, hourly increments are scaled by the CaPA-to-RDPS ratio. When the RDPS 6-hour total falls below this threshold, ratio scaling is not applied; instead, the CaPA 6-hour total is distributed uniformly across the six hours. This fallback prevents unrealistically large scaling factors in dry forecast conditions while preserving the CaPA accumulation.

Figure 1 shows an example daily precipitation field from ORRC v1.0 over the full domain, illustrating the spatial structure resolved at 0.09° resolution.

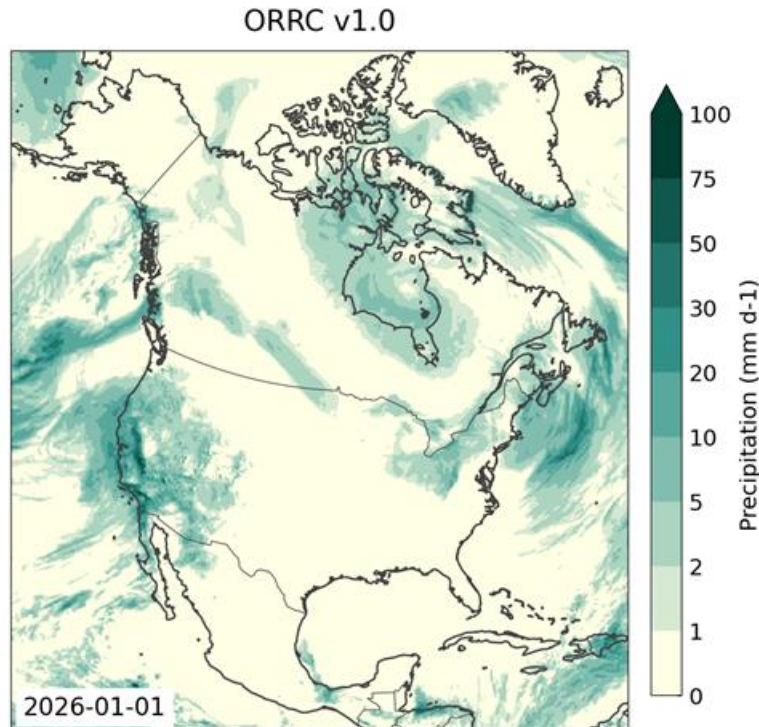


Figure 1. Daily accumulated precipitation (mm day^{-1}) from ORRC v1.0 over North America, Central America, and nearly all of Greenland on 1 January 2026. The map illustrates the spatial structure at 0.09° ($\sim 10 \text{ km}$) resolution on the rotated latitude–longitude grid.

2.3.3 Bias adjustment

To reduce discrepancies relative to the reanalysis data, we produce a bias-adjusted variant of ORRC (ORRC-a) for a subset of daily variables (daily mean precipitation and daily mean/min/max near-surface air temperature) using CaSR v3.2 as a target dataset. For applications requiring closer consistency with CaSR v3.2, we recommend using ORRC-a v1.0 for the variables provided. The same adjustment approach can be applied to other variables if needed. Our main objective of producing bias adjusted fields is to align the near-real-time climate conditions with the historical conditions provided by CaSR, ensuring that year-to-date values are comparable and placed as accurately as possible within the long-term historical context. For applications not requiring comparison within a historical context, the unadjusted ORRC data may be suitable for direct use.

The bias adjustment is implemented using the xsdba Python library (Bourgault et al., 2026). We apply a day-of-year mean bias correction computed using a 31-day moving window (± 15 days). For temperature variables, we apply an additive correction, and for precipitation we apply a multiplicative correction, which maintains non-negative precipitation values and preserves temporal variability around the adjusted mean. Figure 2 illustrates the impact of

the bias adjustment on the monthly mean precipitation time series in a representative region over the overlapping period with CaSR v3.2.

The adjustment period is 2019–2024. The years 2015–2018 are excluded because RDPS is an operational forecasting system whose configuration evolves over time. In our diagnostics, ORRC biases relative to CaSR v3.2 are substantially larger during 2015–2018 than in more recent years and including these earlier years in the adjustment leads to overcorrection.

Figure 3 shows the seasonal mean precipitation bias relative to CaSR v3.2 over the full domain for ORRC v1.0 and ORRC-a v1.0 during the 2019–2024 overlap period. The bias adjustment substantially reduces the magnitude and spatial extent of precipitation biases in all seasons. Additional maps showing the effect of the bias adjustment for near-surface air temperature variables are available in the ORRC GitHub repository at <https://github.com/Ouranosinc/ORRC>.

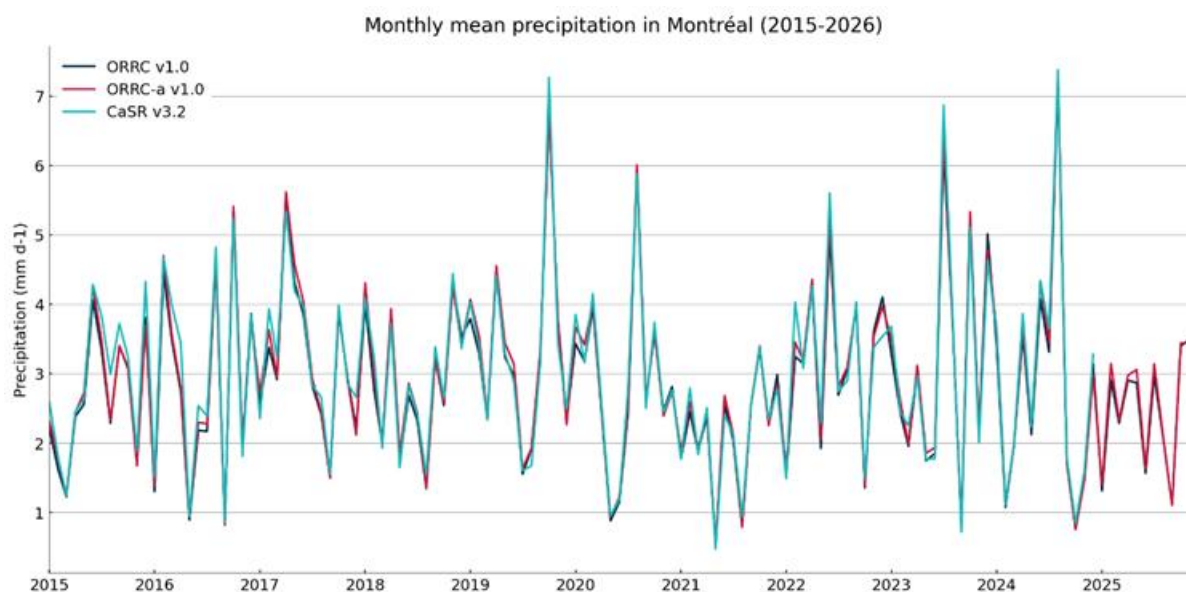


Figure 2. Monthly mean precipitation (mm day^{-1}) in the Montréal regional county municipality for 2015–2026 derived from ORRC v1.0, ORRC-a v1.0, and CaSR v3.2. The bias-adjusted product (ORRC-a v1.0) is designed to reduce mean bias relative to CaSR v3.2 while preserving temporal variability around the mean. CaSR v3.2 is available through 2024, and ORRC v1.0 and ORRC-a v1.0 extend to the present.

Seasonal Bias in Precipitation (2019-2024)

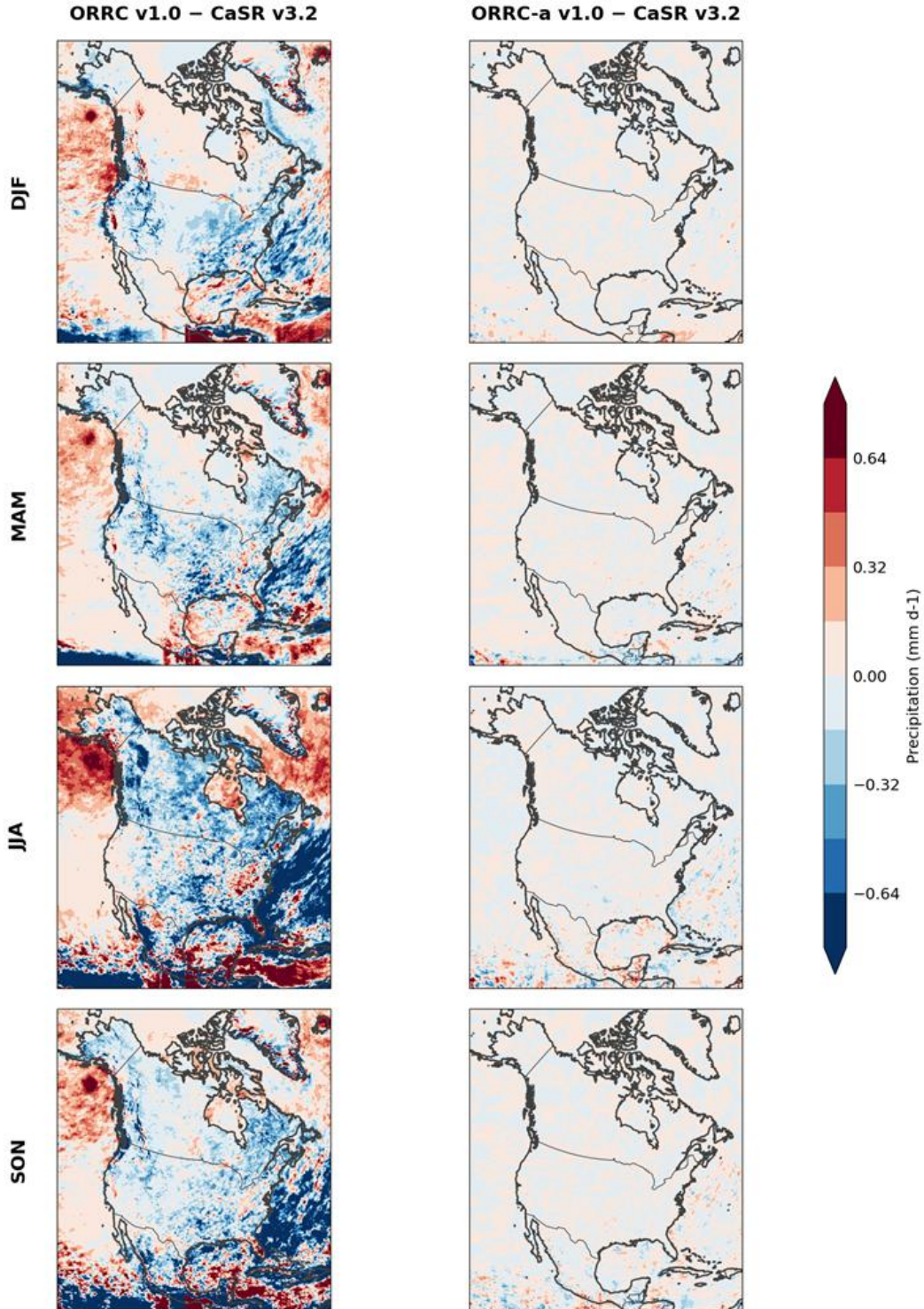


Figure 3. Seasonal mean precipitation bias relative to CaSR v3.2 over 2019–2024 for ORRC v1.0 (left column) and ORRC-a v1.0 (right column), shown for winter (DJF), spring (MAM), summer (JJA), and fall (SON) across North America. Bias is computed as the difference between the seasonal mean precipitation in each ORRC product and the corresponding seasonal mean precipitation in CaSR v3.2.

2.4 Evaluation against CaSR v3.2

We evaluated ORRC v1.0 and ORRC-a v1.0 against CaSR v3.2 over 2019–2024 using a consistent set of diagnostics for all variables, including regional Taylor diagrams and spatial maps of error statistics. The objective of this comparison was to assess the degree of similarity between the products, rather than to identify which product performs better.

For the regional evaluation, diagnostics were summarized over the 31 Bukovsky regions shown in Figure 4 (Bukovsky, 2011). Figure 5 illustrates this evaluation for monthly mean precipitation variability, comparing ORRC v1.0 with CaSR v3.2 across the Bukovsky regions. The diagram shows that ORRC v1.0 captures the regional pattern of precipitation variability reasonably well overall, with generally high correlations, although the normalized standard deviations are often lower than 1, indicating weaker variability than in CaSR v3.2 in many regions. Taylor diagrams for the other variables, as well as for ORRC-a v1.0, are available in the ORRC GitHub repository.

Bukovsky Regions

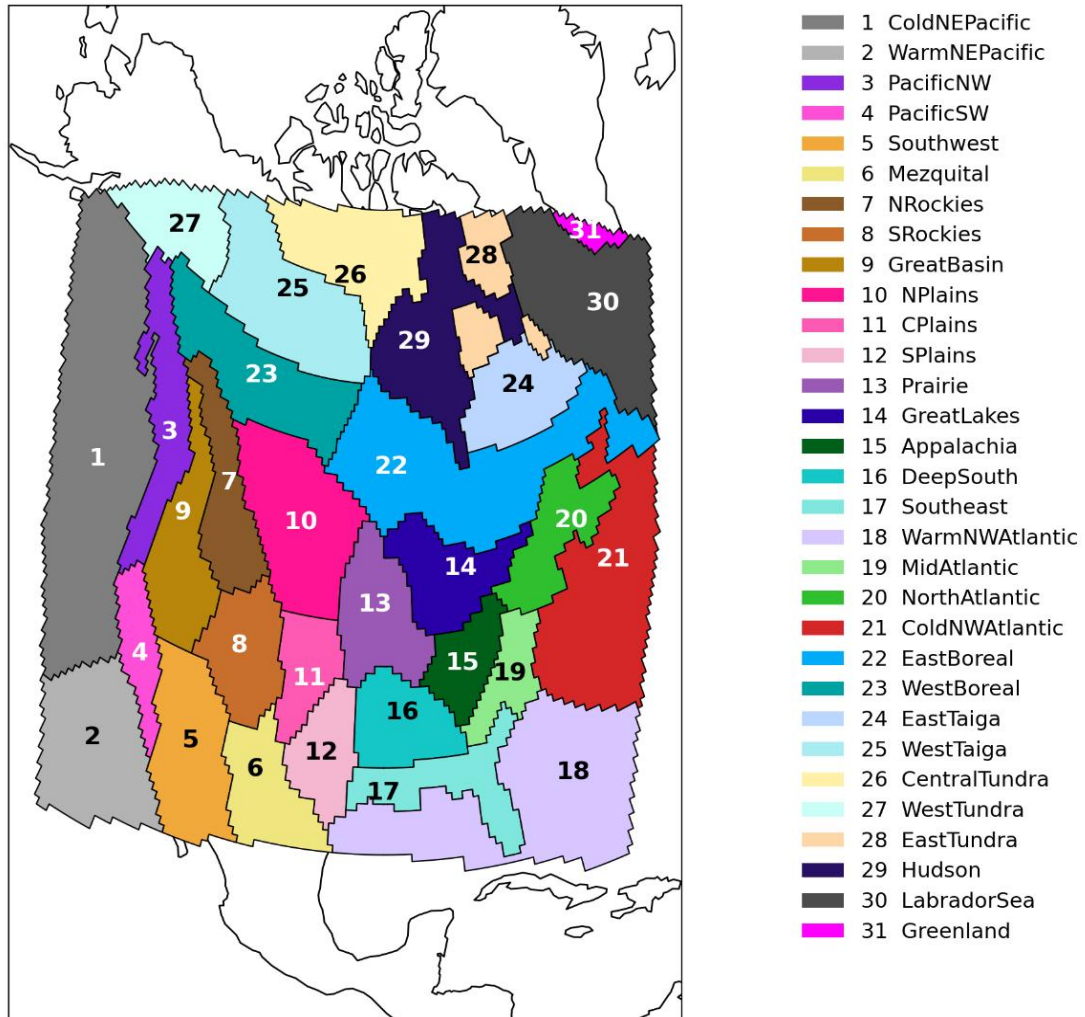


Figure 4. Bukovsky regions used for the regional evaluation of ORRC v1.0 and ORRC-a v1.0 against CaSR v3.2.

Precipitation (2019-2024)

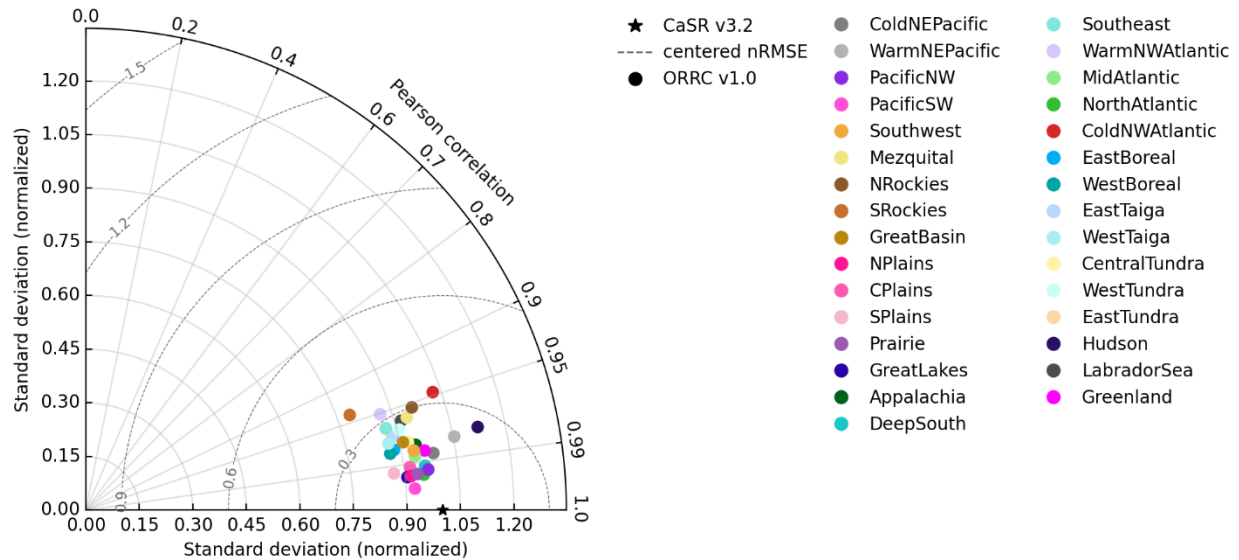


Figure 5. Taylor diagram comparing ORRC v1.0 and CaSR v3.2 monthly mean precipitation over 2019–2024 for the 31 Bukovsky regions. For each region, the statistics are computed from the time series of spatially averaged monthly precipitation. Radial distance indicates the normalized standard deviation, angle indicates the Pearson correlation coefficient and dashed contours show the centered, normalized root-mean-square error (centered nRMSE). The star marks the CaSR v3.2 reference and coloured circles denote the individual Bukovsky regions.

Figure 6 provides a full-domain spatial evaluation of ORRC v1.0 precipitation over 2019–2024 based on seasonal variability in monthly mean precipitation. The left column shows the seasonal standard deviation of monthly mean precipitation in CaSR v3.2, highlighting stronger variability along the Pacific and Atlantic coasts and in lower-latitude regions, and weaker variability over much of the continental interior and the Arctic. The normalized RMSE patterns in the right column show that ORRC v1.0 generally reproduces the broad spatial structure of this variability well, with mostly moderate errors across the domain. Larger normalized errors occur mainly in some coastal and low-variability regions, where modest RMSE values can translate into large normalized values because the reference seasonal standard deviation is small. Grid cells with very low CaSR v3.2 variability are therefore hatched in the nRMSE maps. Corresponding spatial diagnostics for the remaining variables and for ORRC-a v1.0 are available in the ORRC GitHub repository.

Precipitation (2019-2024)

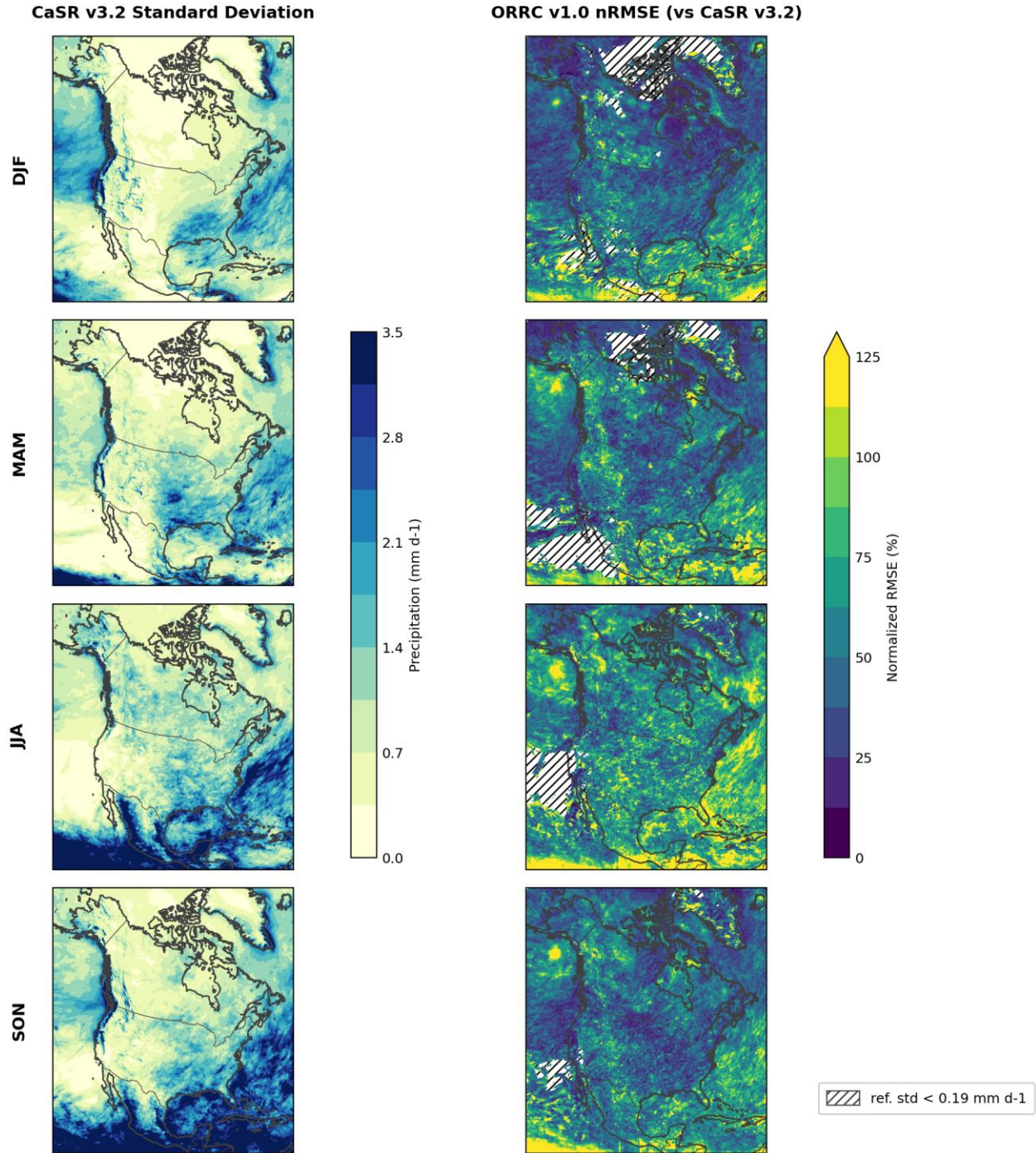


Figure 6. Seasonal standard deviation of monthly mean precipitation from CaSR v3.2 over 2019–2024 (left column) and the corresponding normalized RMSE of ORRC v1.0 relative to CaSR v3.2 (right column) for winter (DJF), spring (MAM), summer (JJA), and fall (SON) across North America. The normalized RMSE is defined as the seasonal RMSE of monthly mean precipitation divided by the seasonal standard deviation of the CaSR v3.2 reference. Grid cells where the CaSR v3.2 seasonal standard deviation is below 0.19 mm/day are hatched, indicating areas of very low reference variability where normalization can yield artificially inflated values. The 0.19 mm/day threshold corresponds to the 5th percentile of positive, finite seasonal standard deviation values of monthly mean CaSR v3.2 precipitation pooled across seasons and grid cells.

References

- Bourgault, P., Dupuis, É., Smith, T. J., Aoun, A., Beaupré Laperrière, A., Braun, M., Gammon, S., Huard, D., Lamarche, A., Lavoie, J., Rondeau-Genesse, G., & Roy, P. (2026). *xsdbs: Statistical downscaling and bias adjustment library* [Computer software]. Zenodo. <https://doi.org/10.5281/zenodo.18613881>
- Bukovsky, M. S. (2011). *Masks for the Bukovsky regionalization of North America*. Regional Integrated Sciences Collective, Institute for Mathematics Applied to Geosciences, National Center for Atmospheric Research. <http://www.narccap.ucar.edu/contrib/bukovsky/>
- dimmilena. (2023). *Dimmilena/rdpa_as_rdrs* [Python]. https://github.com/dimmilena/rdpa_as_rdrs (Original work published 2022)
- Environment and Climate Change Canada. (2025, October 16). *Regional Deterministic Prediction System*. Government of Canada, Open Science and Data Platform. <https://osdp-psdo.canada.ca/dp/en/search/metadata/NRCAN-FGP-1-a9f2828c-0d78-5eb6-a4c7-1fc1219f1e3d>
- Fortin, V., Roy, G., Stadnyk, T., Koenig, K., Gasset, N., & Mahidjiba, A. (2018). Ten Years of Science Based on the Canadian Precipitation Analysis: A CaPA System Overview and Literature Review[†]. *Atmosphere-Ocean*, 56(3), 178–196. <https://doi.org/10.1080/07055900.2018.1474728>
- Gasset, N., Fortin, V., Dimitrijevic, M., Carrera, M., Bilodeau, B., Muncaster, R., Gaborit, É., Roy, G., Pentcheva, N., Bulat, M., Wang, X., Pavlovic, R., Lespinas, F., Khedhaouiria, D., & Mai, J. (2021). A 10 km North American precipitation and land-surface reanalysis based on the GEM atmospheric model. *Hydrology and Earth System Sciences*, 25(9), 4917–4945. <https://doi.org/10.5194/hess-25-4917-2021>
- Lespinas, F., Fortin, V., Roy, G., Rasmussen, P., & Stadnyk, T. (2015). Performance Evaluation of the Canadian Precipitation Analysis (CaPA). *Journal of Hydrometeorology*, 16(5), 2045–2064. <https://doi.org/10.1175/JHM-D-14-0191.1>
- Mai, J., Kornelsen, K. C., Tolson, B. A., Fortin, V., Gasset, N., Bouhemhem, D., Schäfer, D., Leahy, M., Anctil, F., & Coulibaly, P. (2020). The Canadian Surface Prediction Archive (CaSPAr): A Platform to Enhance Environmental Modeling in Canada and Globally. *Bulletin of the American Meteorological Society*, 101(3), E341–E356. <https://doi.org/10.1175/BAMS-D-19-0143.1>